

Chapter 1, Problem 29P

Problem

An electric stove with four burners and an oven is used in preparing a meal as follows.

Burner 1: 20 minutes

Burner 2: 40 minutes

Burner 3: 15 minutes

Burner 4: 45 minutes

Oven: 30 minutes

If each burner is rated at 1.2 kW and the oven at 1.8 kW, and electricity costs 12 cents per kWh, calculate the cost of electricity used in preparing the meal.

Step-by-step solution

Step 1 of 2

$$t_{B1} = 20 \text{ min ;}$$

$$t_{B1} = \frac{1}{3} \text{ hr;}$$

$$t_{B2} = 40 \text{ min;}$$

$$t_{B2} = \frac{2}{3} \text{ hr;}$$

$$t_{B3} = \frac{1}{4} \text{ hr.}$$

$$t_{B4} = \frac{3}{4} \text{ hr ;}$$

$$t_0 = 30 \text{ min}$$

$$t_0 = \frac{1}{2} \text{ hr.}$$

$$t_B = \frac{1}{3} + \frac{2}{3} + \frac{1}{4} + \frac{3}{4} + \frac{1}{2}$$

$$t_B = 2 \text{ hr}$$

$$\text{Energy} = (2 \times 1.2) + \left(\frac{1}{2} \times 1.8 \right) \text{ k W hr}$$

$$\text{Energy} = 2.4 + 0.9$$

$$\text{Energy} = 3.3 \text{ k W hr}$$

Step 2 of 2

$$\text{Cost} = 12 \times 3.3 \text{ cents}$$

$$\boxed{\text{Cost} = 39.6 \text{ cents}}$$